

# llmXive Automated Review of DanceOPD: On-Policy Generative Field Distillation

*llmXive automated review panel*

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**Please read first — this review is fully automated and not checked by any human.** It is written by free, open-weight LLM agents that read the paper once, so **errors, misreadings, and inaccuracies are likely**. Nothing here has been verified by a person. Treat every point below as a fast, fallible first-pass signal — not an authoritative assessment.

## How this review was produced

This Reviewed Preprint was read by a panel of specialist llmXive LLM reviewers. Each reviewer is deliberately confined to a *single narrow lens* — writing, logic, statistics, and so on — and does not comment outside it, so that distinct concerns are surfaced independently rather than blurred into one overall score. The panel runs *once* over the paper. Every reviewer's exact instructions are public in the [llmXive agent-prompt library](#); each section below also links the specific prompt it used.

This report is **advisory, automated feedback for the authors**. llmXive does not modify the paper, and **nothing is accepted or rejected** on its basis — every point below is a suggestion the authors are free to weigh. Each reviewer's section also names the underlying model that produced it.

**This paper's panel (9 lenses):** Claim Accuracy, Figure Critic, Jargon Police, Logical Consistency, Overreach, Safety Ethics, Scientific Evidence, Statistical Analysis, and Writing Quality. Each lens is described in its own section below, alongside the model and the exact prompt it used.

## Claim Accuracy

*Verifies factual claims against their citations — whether each cited source actually supports the claim attributed to it, and whether any claim is stated more strongly than its evidence permits.*

**Reviewer model:** [qwen.qwen3.5-122b](#)

**Prompt:** `agents/prompts/paper_reviewer_claim_accuracy.md`

The review focuses on the accuracy of factual claims and their alignment with the provided data in the tables and text.

The manuscript makes several specific quantitative claims in Section 4.1 (Main Results) regarding performance improvements. A detailed cross-check of these claims against Table 2 (Main Results) and Table 3 (CFG Absorption Diagnostics) reveals minor discrepancies in the reported percentage improvements, which, while small, affect the precision of the factual claims.

1. **GenEval Improvements (Section 4.1):** The text states DanceOPD improves GenEval over the T2I source by 2.0% and over the strongest composition baseline (DiffusionOPD) by 1.6%.
  - **T2I Source:** Table 2 lists T2I at 0.832 and DanceOPD at 0.849. The relative improvement is  $(0.849 - 0.832) / 0.832 \approx 2.04\%$ . The claim of 2.0% is accurate to one decimal place.
  - **DiffusionOPD Baseline:** Table 2 lists DiffusionOPD at 0.833. The relative improvement is  $(0.849 - 0.833) / 0.833 \approx 1.92\%$ . The text claims 1.6%. This is a notable discrepancy (approx. 0.3 percentage points difference). It is possible the authors compared against a different baseline or used a different rounding convention, but as written, the claim “1.6%” is not supported by the numbers in Table 2.
1. **GEditBench Improvements (Section 4.1):** The text claims an 8.5% improvement over the edit source.
  - **Edit Source:** Table 2 lists the Edit source at 4.930 and DanceOPD at 5.347. The relative improvement is  $(5.347 - 4.930) / 4.930 \approx 8.46\%$ . The claim of 8.5% is accurate.
  - **Best Reproduced OPD Baseline:** The text claims an 8.1% improvement over the best reproduced OPD baseline. In Table 2, DiffusionOPD (4.947) is the highest OPD baseline.  $(5.347 - 4.947) / 4.947 \approx 8.09\%$ . This matches the 8.1% claim.
1. **CFG Absorption (Section 4.1):** The text claims a 7.6% improvement over train-only absorption and 1.4% over eval-only CFG.
  - **Train-only:** Table 3 shows Train-only (3.5, 1.0) at 5.422.  $(5.833 - 5.422) / 5.422 \approx 7.58\%$ . Matches 7.6%.
  - **Eval-only:** Table 3 shows Eval-only (1.0, 7.0) at 5.751.  $(5.833 - 5.751) / 5.751 \approx 1.42\%$ . Matches 1.4%.
  - These claims are accurate, but the text would benefit from explicitly stating the  $\alpha$  and  $\beta$  values used for “train-only” and “eval-only” to prevent ambiguity, as the table contains multiple rows.
1. **Ablation Claims (Section 4.2):** The text states hard routing improves over soft mixing by 15.2% (MSE) and 10.6% (KL).

- **MSE:** Table 4 shows Hard Routing (Step alternation, MSE) at 5.751 and Soft-teacher mixing (Soft-teacher MSE) at 4.994.  $\$(5.751 - 4.994) / 4.994 \backslash \text{approx } 15.16\%\$$ . Matches 15.2%.
- **KL:** Table 4 shows Hard Routing (KL) at 5.501 and Soft-teacher mixing (Soft-teacher KL) at 4.976.  $\$(5.501 - 4.976) / 4.976 \backslash \text{approx } 10.55\%\$$ . Matches 10.6%.
- These claims are accurate.

The primary issue is the 1.6% vs 1.92% discrepancy in the GenEval comparison against DiffusionOPD. This suggests a potential calculation error in the manuscript text or a misidentification of the baseline. Given the precision required in scientific claims, this should be corrected to ensure the text accurately reflects the data in Table 2. Additionally, clarifying the specific parameter settings for “train-only” and “eval-only” in the CFG section would strengthen the factual clarity of those claims.

#### Action items.

- **[writing]** In Section 4.1, the claim of a 1.6% GenEval improvement over DiffusionOPD (0.833) is inconsistent with the table data (0.849 vs 0.833 yields  $\sim 1.92\%$ ). Correct the percentage or verify the baseline used.
- **[writing]** In Section 4.1, clarify the specific alpha/beta pairs defining ‘train-only’ and ‘eval-only’ CFG baselines to ensure the 7.6% and 1.4% improvement claims are unambiguously supported by Table 3.
- **[writing]** In Section 4.1, the 8.5% GEditBench improvement over the Edit source (4.930 vs 5.347) is accurate (8.46%), but ensure all percentage claims in the text are rounded consistently to avoid minor discrepancies.

## Figure Critic

*Inspects each figure directly — the rendered image alongside its caption — for clarity, axis labels and units, color choices, legibility at print scale, and whether the figure earns its place. This lens runs on a vision-capable model.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_figure\_critic.md

### Figure 1

The figure effectively visualizes the trade-off between performance and cost, but the caption contains a rendering artifact (‘\$\$’) and the inset legend in the right panel is illegible.

### Figure 2

The figure is a visually rich collage of 28 images, but it fails to communicate its scientific point because it lacks labels or a legend to identify the specific capabilities being demonstrated.

Additionally, the caption contains a grammatical error ('from .') and is too vague to verify the claims of 'diverse behaviors' and 'retained capabilities' without external context.

### Figure 3

The figure provides a clear conceptual illustration of the hard-routing and single-query mechanism with a comprehensive legend, but the caption text is grammatically incomplete.

### Figure 4

The figure effectively demonstrates the method's capability across material, lighting, and style edits, but the caption lacks specific descriptions of the tasks shown. Additionally, the 'Raw' baseline images in the comparison grid do not match the source images in the left column, complicating the visual assessment of content preservation.

### Figure 5

The figure displays a grid of images but fails to label the rows with the corresponding method names (DanceOPD vs. baselines), rendering the caption's comparative claims unverifiable. Additionally, the top row of text labels is cropped.

### Figure 6

The figure contains four subplots but the caption only describes three, leaving the leftmost and rightmost plots unexplained. Additionally, the rightmost plot lacks a legend, and the caption contains a missing mathematical symbol.

### Figure 7

Figure 7 presents qualitative image grids labeled with method names but fails to provide the quantitative evidence or explicit visual annotations required to substantiate the caption's claims about realism-field absorption and comparative effectiveness.

### Figure 8

The figure is critically flawed as the visual content (bar charts with specific model names) does not match the caption's description of diagnostic ablations. Additionally, the legend is cropped and illegible, preventing interpretation of the data.

### Figure 9

The figure displays a grid of images purported to show training progression, but it lacks any labels, axes, or legends to identify the specific steps or iterations shown. Additionally, the visual content does not clearly demonstrate the 'composition' or 'editing' process described in the caption.

## Figure 10

The figure contains a direct contradiction between the rightmost panel’s data and the caption’s claim regarding trajectory queries. Additionally, the first two panels lack legends, rendering the specific ablation series unidentifiable.

## Figure 11

The figure presents a visual comparison of global edits but suffers from a mismatch between the caption’s listed baselines and the grid’s labels, and some method labels are difficult to read.

## Figure 12

The figure provides qualitative comparisons of DanceOPD against baselines for two editing tasks. However, the caption’s reference to ‘Local and Global Edits’ is not visually distinguished in the image labels, and the specific transformations being evaluated are not clearly defined for the reader.

**Action items.**

- **[writing]** Figure 1: The caption contains a rendering artifact ‘Editing\$T2I’ which likely indicates a missing LaTeX symbol (e.g.,  $\rightarrow$ ) between ‘Editing’ and ‘T2I’.
- **[writing]** Figure 1: The inset legend in the right panel (top-left) is illegible; the x-axis label ‘s/step’ is visible, but the y-axis label and tick values are too blurry to read.
- **[writing]** Figure 2: The caption contains a grammatical error and missing reference: ‘Qualitative Examples from .’ ends abruptly with a period and no noun phrase.
- **[science]** Figure 2: The collage contains 28 diverse images but lacks any labels, text overlays, or a legend to indicate which specific ‘text-to-image’ or ‘editing’ behaviors are being demonstrated in each panel.
- **[writing]** Figure 2: The caption claims the figure shows ‘diverse text-to-image and editing behaviors’ but does not specify what the ‘original generative capabilities’ are that are being retained, making the claim unverifiable from the visual alone.
- **[writing]** Figure 3: The caption contains a grammatical omission (‘Conceptual Illustration of .’) and the title ‘DanceOPD’ is missing from the sentence structure, making the description incomplete.
- **[writing]** Figure 4: The caption refers to ‘Additional Edit Cases’ but does not explicitly name the three specific editing tasks shown (Apple material, Piano environment, Armchair lighting), making it harder to map the visual examples to the text description.
- **[writing]** Figure 4: The ‘Raw’ column in the Grand Piano and Leather Armchair rows displays images that do not match the ‘Source’ images in the left column (e.g., the piano is in a different room, the chair is in a different setting), which obscures the baseline for evaluating the editing quality.

- **[science]** Figure 5: The caption claims to show ‘T2I and Edit Composition’ and compares DanceOPD to baselines, but the figure contains no text labels, legends, or row headers to identify which method generated which image. It is impossible to verify the claim that ‘several baselines introduce color shift’ without knowing which rows correspond to the baselines.
- **[writing]** Figure 5: The column headers (e.g., ‘T2I: Buckskin Dusk...’) are cropped at the top edge of the image, making the specific prompts used for generation partially illegible.
- **[science]** Figure 6: The caption describes three panels (a, b, c), but the rendered image contains four distinct subplots. The leftmost subplot (T2I Score vs Step) is not referenced in the caption, and the rightmost subplot (GEditBench vs Step) is not described, creating a disconnect between the text and the visual evidence.
- **[writing]** Figure 6: The caption mentions ‘where  $\$$  denotes the effective CFG’, but the mathematical symbol for the effective CFG is missing from the text, making the sentence incomplete and confusing.
- **[science]** Figure 6: The rightmost subplot (GEditBench-EN Avg) lacks a legend defining the specific line styles (solid vs dashed) and colors for the ‘Gen-8’, ‘Gen-16’, etc. series, rendering the data uninterpretable without guessing.
- **[science]** Figure 7: The caption claims to show ‘Realism-Field Absorption’ and compares DanceOPD to off-policy distillation, but the image is a grid of qualitative samples labeled ‘Base’, ‘Teacher’, ‘Off-Policy’, and ‘Ours’ without any quantitative metrics, statistical significance indicators, or explicit visual markers demonstrating the claimed ‘shift toward more photorealistic texture’.
- **[writing]** Figure 7: The caption states ‘DanceOPD absorbs the teacher’s realism-oriented field more effectively than off-policy distillation’, but the figure lacks a clear legend or annotation explaining what specific visual features constitute ‘realism’ or how the comparison is quantified beyond subjective visual inspection.
- **[fatal]** Figure 8: The rendered image contains three bar charts with x-axis labels (e.g., ‘K1G1’, ‘Hard MSE’, ‘MSE’) that do not correspond to the caption’s description of ‘Routing, Objective, and Dense-Query Diagnostics’ (e.g., ‘same-step accumulation’, ‘soft teacher mixing’). The figure appears to be a mismatch or placeholder.
- **[fatal]** Figure 8: The legend at the bottom left is cropped and illegible, making it impossible to map the bar colors to the specific experimental conditions or baselines mentioned in the caption.
- **[science]** Figure 8: The y-axis label ‘GEditBench-EN Avg’ is present, but the specific metric or task (e.g., ‘Global Edit’, ‘Local Edit’) is not defined in the caption or axis, making the performance claims ambiguous.
- **[science]** Figure 9: The caption describes ‘Training Progression’ and ‘distillation proceeds’, but the image contains no text labels, axis markers, or legends to indicate which column corresponds to which training step or iteration.

- **[science]** Figure 9: The caption claims the figure shows ‘Local and Global Edit Composition’, yet the displayed rows (cars, landscapes, bikes, flowers, headphones, drinks) appear to be independent single-image generations rather than explicit composition tasks or edits of a base image.
- **[science]** Figure 10: The rightmost panel shows ‘Weighted’ performance increasing with Trajectory Queries  $K$ , contradicting the caption’s claim that ‘increasing the number of trajectory queries does not improve performance’.
- **[writing]** Figure 10: The left and middle panels lack legends defining the ‘Low-t’, ‘Median-t’, ‘High-t’, ‘Local’, ‘T2I’, ‘Merged’, and ‘Global’ series, making the ablation trends uninterpretable without guessing.
- **[science]** Figure 11: The caption claims to compare DanceOPD with ‘off-policy distillation, joint training, DiffusionOPD, and Flow-OPD’, but the figure grid only displays ‘Raw’, ‘Off-Policy’, ‘FlowOPD’, ‘Teacher’, ‘Joint Training’, and ‘DiffusionOPD’. The ‘DanceOPD (Ours)’ result is shown as a large standalone image on the left, but the specific ‘Off-Policy’ baseline shown in the grid is not explicitly labeled as ‘Off-Policy Distillation’ in the caption context, and the ‘Teacher’ baseline is included in th
- **[writing]** Figure 11: The ‘Raw’ and ‘Off-Policy’ labels in the top row are partially obscured by the image content or have low contrast against the background, reducing legibility.
- **[writing]** Figure 12: The caption refers to ‘Local and Global Edits’ but the image labels are ‘Evening Gown’ and ‘Rental Room’, which do not explicitly indicate the edit type (local vs global) for the reader.
- **[writing]** Figure 12: The caption claims DanceOPD produces ‘stronger global transformations’ than baselines, but the ‘Rental Room’ example shows DanceOPD (Ours) as a clean render while baselines like ‘Raw’ and ‘Off-Policy’ show messy or cluttered scenes, making the comparison ambiguous without a clear ‘before’ state or explicit transformation description.

## Jargon Police

*Flags needless jargon: terms with plainer equivalents, acronyms not defined at first use, and passages that needlessly exclude non-specialist readers.*

**Reviewer model:** `qwen.qwen3.5-122b`

**Prompt:** `agents/prompts/paper_reviewer_jargon_police.md`

The manuscript relies heavily on specialized terminology that is not consistently defined for a broader audience, creating barriers for non-specialist readers. The term “semantic-side” (Section 3.3, Eq. 4) is used frequently to describe the query distribution but is never explicitly defined; readers must infer it means “low-noise” or “high-fidelity” regions of the trajectory. Similarly, “anchor capability” (Introduction, Section 4.1) is used to denote the preserved base capability (e.g., T2I) without a clear definition.

The concept of a “rollout” (Section 3.2, Eq. 3) is borrowed from reinforcement learning but

applied to flow matching; while the equation shows the process, a brief textual explanation of what a rollout represents in this specific generative context would aid clarity. The use of “stop-gradient” (Section 3.2) is also introduced with notation `sg` but lacks a plain-language explanation of its purpose (preventing backpropagation through the solver). Finally, the core concept of a “capability field” (Section 3.1) is introduced as a velocity field but could benefit from a simpler introductory sentence explaining that it represents the direction of change for a specific task. Defining these terms upon first use would significantly improve accessibility without sacrificing technical precision.

#### Action items.

- **[writing]** The manuscript relies heavily on specialized terminology that is not consistently defined for a broader audience, creating barriers for non-specialist readers. The term “semantic-side” (Section 3.3, Eq. 4) is used frequently to describe the query distribution but is never explicitly defined; readers must infer it means “low-noise” or “high-fidelity” regions of the trajectory. Similarly, “anchor capability” (Introduction, Section 4.1) is used to denote the preserved base capability (e.g., T2I) wi

## Logical Consistency

*Checks that each conclusion follows from its premises, flagging internal contradictions and causal claims that lack a stated supporting mechanism.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_logical\_consistency.md

The logical consistency of the proposed method relies heavily on the equivalence between the proposed MSE objective and a KL-divergence minimization, as well as the linearity of the guidance field composition.

First, in Section 3.4 and Appendix A.1, the authors derive that minimizing the KL divergence between the student and teacher transition kernels reduces to a velocity MSE objective. This derivation explicitly assumes that the student and teacher share the **same local covariance** ( $\sigma_t^2$ ). While this is a standard assumption in flow matching, the paper does not logically justify why the student’s on-policy rollout distribution (which is being actively trained) would maintain the same local covariance structure as the frozen teacher field, especially given that the student is learning to compose conflicting capabilities. If the student’s local variance differs significantly from the teacher’s, the MSE objective is no longer the exact maximum likelihood estimator for the KL divergence, introducing a potential logical gap between the theoretical motivation and the practical objective.

Second, regarding the CFG absorption results in Section 4.1.D, the paper argues that the effective guidance strength is the product  $\alpha\beta$  (Eq. 14 in Appendix A.8). This logic holds strictly only if the student  $v_\theta$  perfectly approximates the guided teacher field  $v_\alpha$  (i.e.,  $v_\theta \approx v_\alpha$ ). The paper presents empirical evidence of “over-guidance” when  $\alpha\beta$  is large but does not provide a logical bound or analysis of how approximation errors in  $v_\theta$  propagate through the composition  $v_{\{\emptyset\}}$

+  $\beta(v_{\theta} - v_{\emptyset})$ . If the student fails to perfectly absorb the teacher, the resulting field is not a simple affine scaling, and the claim that the performance drop is solely due to the  $\alpha\beta$  magnitude is an oversimplification that lacks rigorous support in the text.

Finally, the argument that “dense trajectory queries” fail due to “trajectory-query correlation” (Sec 3.3) is logically sound in principle, but the paper assumes that the correlation  $\rho$  is high enough to negate the variance reduction of averaging  $K$  samples. While the SDE decorrelation experiment supports this, the paper does not explicitly quantify the correlation coefficient  $\rho$  in the ODE rollout case to prove that the variance reduction factor  $1 + (K-1)\rho$  is indeed close to  $K$  (i.e.,  $\rho \approx 1$ ). Without this quantitative link, the leap from “states are correlated” to “dense queries are strictly worse than single queries” remains a qualitative assertion rather than a derived necessity.

#### Action items.

- **[writing]** The logical consistency of the proposed method relies heavily on the equivalence between the proposed MSE objective and a KL-divergence minimization, as well as the linearity of the guidance field composition. First, in Section 3.4 and Appendix A.1, the authors derive that minimizing the KL divergence between the student and teacher transition kernels reduces to a velocity MSE objective. This derivation explicitly assumes that the student and teacher share the same local covariance ( $\sigma_t^2$

## Overreach

*Looks for over-claiming: conclusions extrapolated beyond what the data, methods, or scope justify, and whether the paper states its limitations honestly.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_overreach.md

The paper makes several claims that extend beyond the immediate evidence provided in the experiments, particularly regarding the magnitude of performance gains over specialized teachers and the generality of the “absorption” capability.

First, in the Introduction (lines 105-108), the authors state that DanceOPD improves GEditBench “over the edit source by 8.5%.” This is a significant over-claim. In a distillation setting, a student model trained to mimic a frozen teacher (the “edit source”) should theoretically converge to the teacher’s performance, not exceed it, unless the student has access to a larger or different dataset than the teacher, or the teacher was evaluated under different conditions. The text does not clarify if the “edit source” baseline was re-evaluated with the same inference settings or if the student benefited from data augmentation not available to the teacher. Without this clarification, the claim implies the student has learned a “super-teacher” capability, which contradicts the standard definition of distillation and suggests an over-interpretation of the results.

Second, the Abstract and Section 3 claim that the framework “absorbs operator-defined fields such as classifier-free guidance.” While Section 4.1.D demonstrates that the student can learn a CFG-guided field, the results explicitly show that combining the absorbed field with

external inference-time CFG leads to “over-guided composition” that “drops substantially” (lines 138-140). The paper frames this as a successful absorption, but the data indicates a narrow operating window where this works; the claim should be tempered to reflect that it internalizes the field but requires careful inference-time scaling to avoid degradation.

Finally, the Conclusion (lines 230-232) asserts that the work establishes a “practical route for generative field distillation in flow-matching models” and suggests it is a path toward “scalable multi-capability visual generation.” The experiments are confined to two specific composition settings (T2I+Edit, Local+Global) and two base models. There is no evidence provided regarding the scalability of the method to larger model sizes, more complex capabilities (e.g., 3D, video, or complex reasoning tasks), or a wider variety of conflicting tasks. The claim of “scalable” is an extrapolation not supported by the current experimental scope. The conclusion should be narrowed to reflect the specific capabilities and models tested.

#### Action items.

- **[science]** Claiming the student outperforms the ‘edit source’ teacher by 8.5% (Sec 1) is an over-claim. Distillation should not exceed the teacher without extra data or different eval protocols. Clarify if the teacher was re-evaluated under identical conditions or if this implies a super-teacher capability, which contradicts standard distillation.
- **[writing]** The claim that the method ‘absorbs’ CFG (Abstract, Sec 3) overstates the results. Sec 4.1.D shows absorbed and external CFG compose multiplicatively, causing severe degradation if not carefully managed. Temper the claim to reflect that it internalizes the field but requires strict inference-time scaling to avoid over-guidance.
- **[writing]** The conclusion that the method is a ‘practical route toward scalable multi-capability visual generation’ (Sec 5) is too broad. Experiments are limited to two composition settings and specific models. There is no evidence for scalability to larger models or complex tasks. Restrict the conclusion to the demonstrated capabilities.

## Safety Ethics

*Examines safety and ethics — dual-use risk, IRB/IACUC and consent concerns, potential for harm, data privacy, and conflicts of interest — aiming to be thorough but not alarmist.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_safety\_ethics.md

The manuscript presents a technical framework for multi-capability image generation but lacks sufficient detail regarding the safety and ethical implications of the data and models involved.

First, the evaluation of the “realism-field absorption” setting relies on a **proprietary realism reward model** (Section 4.1.C, Appendix “Other Experiments Details”). The authors state, “Our reward model is proprietary, which assigns a photorealism score to each input image.” There is no information regarding the training data of this reward model, its potential biases (e.g., towards specific skin tones, body types, or cultural aesthetics), or whether it has been audited for safety. If the reward model is biased, the distillation process could amplify these

biases in the student model. The authors must provide a description of the reward model’s provenance and a discussion of how potential biases were mitigated.

Second, the training and evaluation involve image editing datasets (e.g., GEditBench, OmniEdit) and the generation of edited images. The manuscript does not explicitly state whether the training data was screened for **sensitive or harmful content** (e.g., non-consensual deepfakes, hate symbols, sexually explicit material, or depictions of violence). Similarly, there is no mention of safety filters applied to the generated outputs. Given the dual-use nature of image editing models, a statement on data curation protocols and the implementation of content safety filters is required.

Finally, the “CFG Absorption” technique (Section 4.1.D) internalizes the classifier-free guidance mechanism into the student model. While the paper frames this as a performance optimization, it raises a safety concern: if the base model relies on inference-time guidance to enforce safety constraints (e.g., refusing to generate harmful content), absorbing this guidance could potentially **bypass those safety mechanisms**. The authors should address whether their method preserves the safety constraints of the underlying foundation model or if there is a risk of generating unsafe content that the original model would have filtered.

#### Action items.

- **[writing]** The paper relies on a proprietary realism reward model for evaluation (Sec 4.1.C, App. Settings) without describing its training data, potential biases, or safety guardrails. A description of the reward model’s provenance and a discussion of potential failure modes (e.g., bias towards specific demographics or unsafe content) is required to assess the safety of the generated outputs.
- **[writing]** The method involves training on image editing datasets (e.g., GEditBench, OmniEdit) and generating edited images. The manuscript does not explicitly state whether the training data or the generated outputs were screened for sensitive content (e.g., deepfakes, non-consensual imagery, hate symbols). A statement on data curation and content safety filters is necessary.
- **[writing]** The paper discusses “CFG Absorption” (Sec 4.1.D), which internalizes guidance scales. While primarily a performance optimization, this could theoretically be used to bypass safety filters embedded in the inference-time guidance of the base model. The authors should address whether their method preserves or inadvertently bypasses safety constraints of the underlying foundation model.

## Scientific Evidence

*Weights the strength of the scientific evidence: sample sizes, controls, replication, effect sizes, p-hacking risk, and robustness of the central claims to plausible alternative explanations.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_scientific\_evidence.md

The scientific evidence presented in the paper is generally strong in its experimental design,

particularly in the systematic ablation of the three core design choices (routing, query state, and query density). The use of multiple baselines (Joint Training, Weight Merging, Off-Policy, DiffusionOPD, Flow-OPD) across four distinct settings (T2I/Edit, Local/Global, Realism, CFG) provides a comprehensive view of the method’s performance. The ablation studies in Section 4.2 and 4.3 effectively isolate the contribution of each component, supporting the claim that hard routing and single-query on-policy distillation are superior to soft mixing and dense querying.

However, the statistical rigor of the reported results requires strengthening. The main results in Table 2 and the ablation tables (Tables 4, 5, 6) report single-point averages without any measure of variance (standard deviation or standard error). In deep learning experiments, especially those involving stochastic training and evaluation on benchmarks like GEditBench and GenEval, performance can vary significantly across different random seeds. The claimed improvements (e.g., 8.1% over the best OPD baseline, 16.1% over the best composition baseline) are substantial, but without confidence intervals or a report of the number of independent runs (e.g., “mean  $\pm$  std over 3 seeds”), it is difficult to assess the statistical significance of these gains. A p-value or a statement that the improvement exceeds the variance of the baseline is necessary to rule out the possibility that the results are due to favorable random seeds.

Furthermore, the evaluation of the “Realism Absorption” setting relies entirely on a “proprietary” reward model (Section 4.1.C and Appendix). While the authors claim this model is used for evaluation, the lack of transparency regarding its architecture, training data, and validation against human preference makes it impossible to independently verify the claim that DanceOPD “closes 85.3% of the student-to-teacher reward gap.” If the reward model is biased or poorly calibrated, the entire conclusion regarding realism absorption is unsupported. The authors should either release the reward model (or a proxy) or provide a detailed technical report on its construction and validation to ensure the evidence is reproducible.

Finally, the baseline “DiffusionOPD” is a reproduction implemented by the authors (Appendix A.2). While the authors describe the reproduction details, there is no evidence provided that this reproduction was first validated against the original DiffusionOPD paper’s results on a standard benchmark to ensure parity. If the reproduction is suboptimal, the reported 8.1% improvement could be an artifact of a weak baseline rather than a genuine advantage of DanceOPD. A validation step showing that the reproduced DiffusionOPD matches the original paper’s reported performance on a shared subset of data is required to establish the validity of the comparison.

#### **Action items.**

- **[science]** The paper claims significant improvements (e.g., 8.1% on GEditBench) but does not report standard deviation, confidence intervals, or the number of independent training runs. Without variance estimates or a statistical significance test (e.g., t-test), it is impossible to determine if these gains are robust or due to random seed variance.
- **[science]** The ‘DiffusionOPD’ baseline is described as a ‘paper-faithful reproduction’ in the appendix. The review requires explicit confirmation that the reproduction code was validated against the original paper’s reported numbers on a shared subset before being used as a baseline for comparison, to ensure the 8.1% gain is not an artifact of a weak baseline implementation.

- **[science]** The ‘Realism Absorption’ experiment relies on a ‘proprietary’ reward model for evaluation. The paper must provide a detailed description of this model’s architecture, training data, and validation metrics, or release a proxy version, to allow independent verification of the claimed 9.9% improvement and the 85.3% gap closure.

## Statistical Analysis

*Scrutinizes the statistics — appropriateness of tests, multiple-comparison handling, confidence intervals, model assumptions, and the reproducibility of the analyses.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_statistical\_analysis.md

The statistical reporting in the experimental section requires significant strengthening to support the paper’s quantitative claims. While the experimental design (ablation of routing, query density, and objectives) is logically sound, the presentation of results lacks the necessary statistical rigor to validate the reported improvements.

**1. Lack of Variance Reporting:** Throughout Section 4 and Tables 2, 4, 5, and 6, results are presented as single-point averages (e.g., “GEditBench average over the best reproduced OPD baseline by 8.1%”). There is no mention of standard deviation, standard error, or confidence intervals. In generative modeling, metrics like GEditBench and GenEval can exhibit variance depending on the random seed used for the evaluation set or the generation process itself. Without error bars or variance metrics, it is impossible to determine if the reported gains (e.g., the 16.1% improvement in local/global edit composition) are statistically significant or within the noise floor of the evaluation protocol.

**2. Missing Replication Details:** The manuscript does not explicitly state the number of random seeds used for training or evaluation. For instance, in Section 4.3 (Ablation Study), the comparison between “Low-t” and “Median-t” queries shows a 23.7% difference. Without knowing if these results are averaged over multiple seeds (e.g., 3, 5, or 10) or represent a single run, the reliability of the “semantic-side single query” conclusion is compromised. The authors should explicitly state the number of seeds and report the mean  $\pm$  standard deviation for all key metrics.

**3. Statistical Significance of Claims:** In Section 4.2, the paper claims that “same-step accumulation... loses 4.6% on average” and that “dense-query... drops by 22.8%.” These are strong claims about the failure modes of specific design choices. To support these, the authors should perform statistical significance testing (e.g., paired t-tests or Wilcoxon signed-rank tests) comparing the proposed method against the ablated variants. The current text relies solely on the magnitude of the difference, which is insufficient for a rigorous statistical analysis.

**4. Evaluation Protocol Consistency:** While the evaluation metrics (GEditBench, GenEval) are standard, the statistical stability of these benchmarks is not discussed. If the evaluation set is small or the metric is sensitive to specific prompt variations, the reported “averages” might be unstable. The authors should clarify if the evaluation sets were fixed across all runs or if they were resampled, and how this impacts the variance of the reported scores.

In summary, the paper needs to include variance metrics (std dev/CI) for all tables, specify the number of random seeds used, and provide statistical tests for the key comparative claims to ensure the conclusions are robust.

#### Action items.

- **[science]** Report standard deviations or confidence intervals for all quantitative results in Tables 2, 4, 5, and 6. The current tables present single-point averages (e.g., GEditBench Avg) without indicating variance across seeds or runs, making it impossible to assess statistical significance of the reported improvements (e.g., the 8.1% gain over DiffusionOPD).
- **[science]** Clarify the number of random seeds used for all experiments. The text mentions ‘reproduced baselines’ and ‘diagnostic studies’ but does not specify if the reported metrics are means over multiple independent runs or single runs. Without this, the robustness of the ‘single-query’ vs ‘dense-query’ conclusions cannot be statistically verified.
- **[science]** In Section 4.2 (Multi-Teacher Extension), the claim that ‘same-step accumulation... loses 4.6% on average’ lacks a statistical test. Please provide p-values or effect sizes to confirm that the observed degradation is statistically significant and not due to random variance in the evaluation metrics.

## Writing Quality

*Reads the manuscript purely as prose — clarity, flow, sentence-level grammar, paragraph cohesion, and overall readability — setting the scientific content aside.*

**Reviewer model:** qwen.qwen3.5-122b

**Prompt:** agents/prompts/paper\_reviewer\_writing\_quality.md

The manuscript demonstrates a high standard of technical writing, with a clear logical progression from problem definition to method formulation and experimental validation. The prose is generally precise, and the mathematical notation is consistent throughout. The introduction effectively sets the stage by clearly articulating the three core alignment challenges (target-field ambiguity, state-distribution mismatch, and trajectory-query correlation), which serves as a strong organizational anchor for the rest of the paper.

However, there are specific areas where the flow and readability can be improved. In Section 4.1, the paragraph describing the T2I and Edit Composition results contains a repetitive list of figure references (“Fig. 5, Fig. 6, Fig. 7, Fig. 8, Fig. 9, and Fig. 10 show the same pattern...”). This enumeration breaks the narrative flow and feels redundant. It would be more effective to state that “Qualitative results across these settings (Figs. 5–10) consistently demonstrate...” or to consolidate the references if the figures are indeed showing identical patterns.

Additionally, the term “semantic-side” is central to the method’s design (specifically regarding the query distribution  $q_{\{\mathrm{sem}\}}$ ) but is introduced with minimal intuitive context in the main text. While the appendix provides the mathematical definition (Beta distribution over rollout coordinates), the main text relies heavily on this term without a brief, plain-language explanation of what “semantic-side” implies for the image generation process (e.g., “the low-noise

region where semantic details are concentrated”). Adding a clarifying clause in Section 3.3 or the Introduction would make the text more accessible to a broader audience.

Finally, the transition between the subsections in Section 4.2 (“Multi-Teacher Extension”) is slightly abrupt. The shift from discussing “Hard Routing” directly to “Same Step Multi-Teacher Accumulation” lacks a connecting sentence that explains why these two specific diagnostics are being paired. A brief sentence linking the sample-level routing decision to the optimizer-level accumulation strategy would enhance the logical cohesion of the experimental analysis.

Overall, the writing is strong and the arguments are well-supported, but these minor adjustments to flow and terminology clarity would further polish the manuscript.

**Action items.**

- **[writing]** In Section 4.1 (Main Results), the sentence listing Figure references (‘Fig. 5, Fig. 6...’) is repetitive and disrupts the flow. Consolidate this list or refer to a single composite figure if the content is identical across them.
- **[writing]** The phrase ‘semantic-side’ is used frequently (e.g., Sec 3.3, 4.3) without a clear, concise definition in the introduction. While defined technically in the appendix, a brief intuitive explanation in the main text would improve accessibility for readers unfamiliar with the specific rollout coordinate terminology.
- **[writing]** In Section 4.2, the transition between the ‘Hard Routing’ and ‘Same Step Multi-Teacher Accumulation’ subsections is abrupt. A brief bridging sentence explaining how these two diagnostics relate to the broader ‘Multi-Teacher Extension’ goal would improve cohesion.